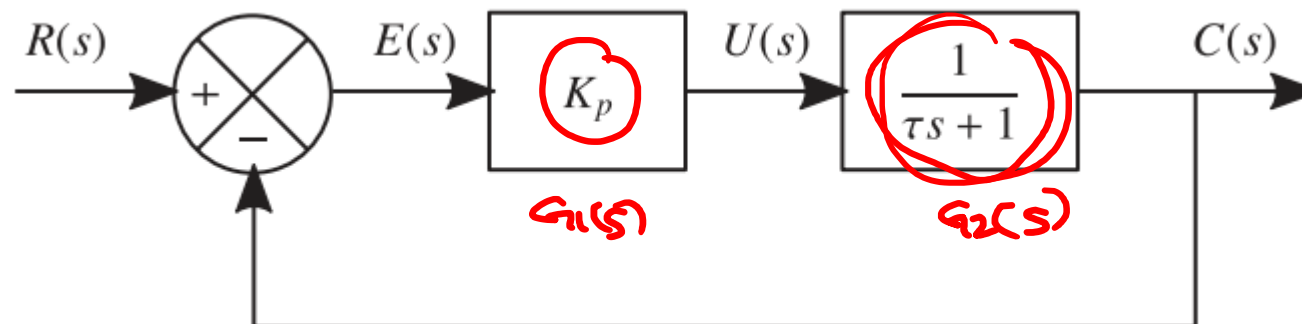


Proportional Control of a First-Order System



$$\frac{C(s)}{R(s)}$$

$$\frac{E(s)}{R(s)}$$

$$R(s) = \frac{1}{s}$$

$$\frac{C(s)}{R(s)} = \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s)} = \frac{K_p / (s\tau + 1)}{1 + K_p / (s\tau + 1)} = \frac{K_p}{s\tau + 1 + K_p}$$

$$C(s) = \frac{1}{s} \times \frac{K_p}{s\tau + 1 + K_p} =$$

$$\frac{K_p}{s(s\tau + 1 + K_p)}$$

$$C(t)$$

$$C_{ss} = \lim_{s \rightarrow 0} s C(s) = \lim_{s \rightarrow 0} s \frac{K_p}{s(s\tau + 1 + K_p)} = \frac{K_p}{K_p + 1}$$

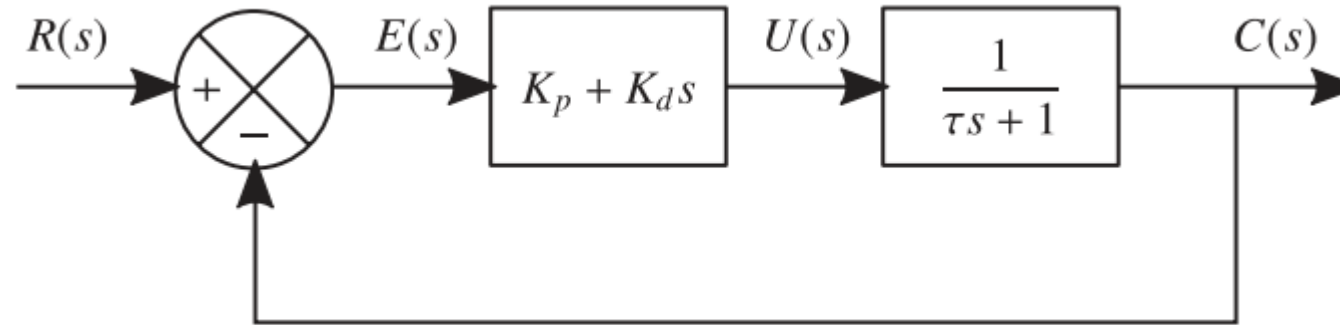
$$\frac{E(s)}{R(s)} = \frac{1}{1 + G_1(s)G_2(s)} = \frac{1}{1 + K_p(zs+1)} = \frac{zs+1}{zs+1+K_p}$$

$$R(s) = \frac{1}{s} \quad ; \quad E(s) = \frac{1}{s} \left(\frac{zs+1}{zs+1+K_p} \right)$$

$$E(s) = \frac{zs+1}{s(zs+1+K_p)}$$

$$P_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \cdot \frac{zs+1}{s(zs+1+K_p)} = \frac{1}{1+K_p}$$

Proportional Plus Derivative Control of a First-Order System



$$\frac{C(s)}{R(s)}, \frac{E(s)}{R(s)}$$

$$C_{ss}, E_{ss}$$

$$R(s) = 1/s$$

$$\frac{C(s)}{R(s)} = \frac{K_p + K_d s}{(\tau + K_d)s + 1 + K_p}$$

$$C(s) = \frac{K_p + K_d s}{s[(\tau + K_d)s + 1 + K_p]}$$

$$C_{ss} = \lim_{s \rightarrow 0} s \frac{K_p + K_d s}{s[(\tau + K_d)s + 1 + K_p]} = \frac{K_p}{K_p + 1}$$

$$\frac{E(s)}{R(s)} = \frac{\tau s + 1}{(\tau + k_d)s + 1 + k_p}$$

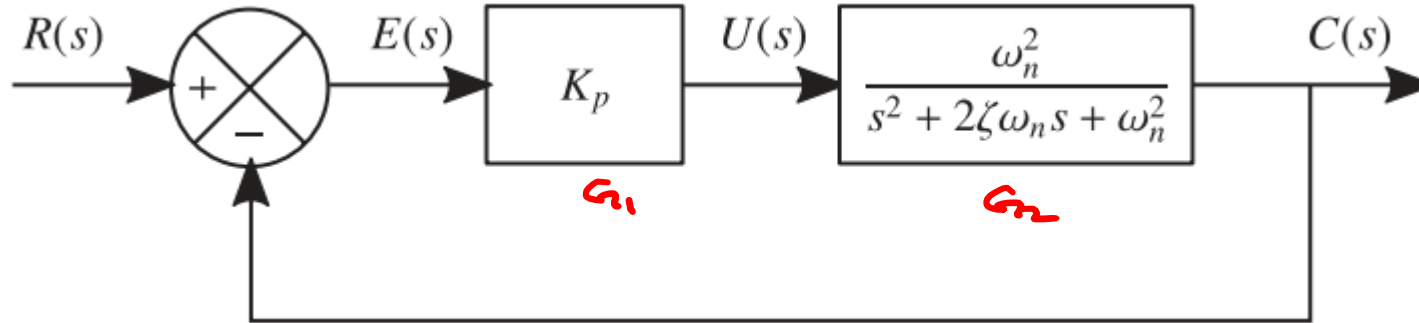
$$E(s) = \frac{\tau s + 1}{s[(\tau + k_d)s + 1 + k_p]}$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \frac{\tau s + 1}{s[(\tau + k_d)s + 1 + k_p]}$$

$\nearrow 0$
 $\searrow 0$

$$e_{ss} = \frac{1}{k_p + 1}$$

Proportional Control of a Second-Order System



CCS

$$C_{ss} = \frac{K_p}{K_p + 1}$$

$$e_{ss} = \frac{1}{K_p + 1}$$

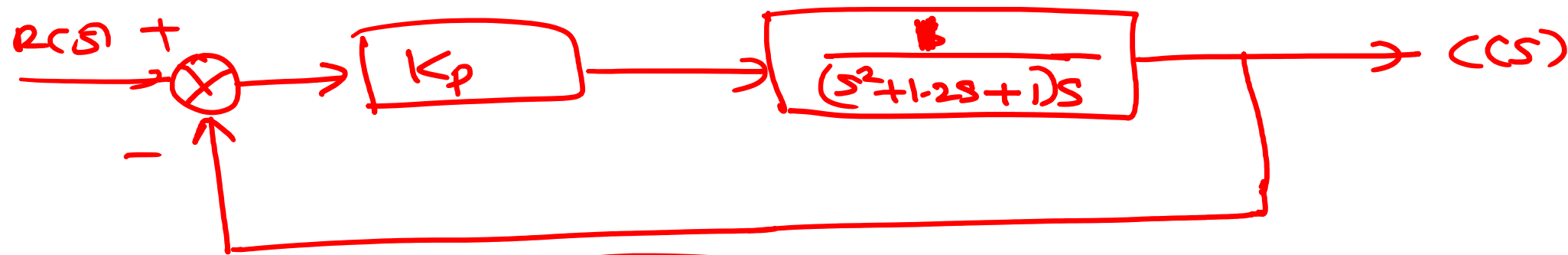
Exercise

A plant with transfer function $G(s)$ is controlled by a controller of variable proportional gain K_p and unity negative feedback. Given

$$G(s) = \frac{1}{s(s^2 + 1.2s + 1)}$$

Show that the value of the proportional gain K_p has no influence on the steady state value of the response of the plant to a unit step input. What is the effect on the stability of the system of a negative value of K_p ?

Investigate the stability of the system for the values: $K_p = 1; 1.5$



✓ $\frac{C(s)}{R(s)} = \frac{K_p}{s(s^2 + 1.2s + 1) + K_p}$ → $\frac{C(s)}{R(s)}$ → poles

$R(s) = \frac{1}{s} \Rightarrow C(s) = \frac{K_p}{s(s^3 + 1.2s^2 + s + K_p)}$

$\lim_{s \rightarrow 0} s \cdot \frac{K_p}{s(s^3 + 1.2s^2 + s + K_p)} = \frac{K_p}{K_p} = 1$

↓ 0